

Sparse Combs Give Infinite Weak Riesz Norm for Every $0 < \beta < 1$

Run `fa_banach_001`, agent `agent_lane_05`

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Abstract

We give a claimed affirmative answer to Open Problem 1.6 in Spector's *A Noninequality for the Fractional Gradient*. For every $d \geq 2$ and $0 < \beta < 1$, we construct a nonnegative $u \in L^1(\mathcal{H}_\infty^{d-\beta})$ such that

$$\sup_{t>0} t \mathcal{H}_\infty^{d-\beta}(\{|DI_1 u| > t\}) = +\infty.$$

The construction uses sparse one-dimensional combs, a fixed tangential cube, and a summable gluing of widely separated smooth blocks.

Source Question

Spector asks on page 6, Open Problem 1.6 of [1]:

For $\beta \in (0, 1)$, can one find $u \in L^1(\mathcal{H}_\infty^{d-\beta})$ such that

$$\sup_{t>0} t \mathcal{H}_\infty^{d-\beta}(\{|DI_1 u| > t\}) = +\infty?$$

$$(1.15) \quad \mathcal{H}_\infty^{d-\beta}(\{|Ru| > t\}) \leq \frac{C}{t} \int_0^\infty \mathcal{H}_\infty^{d-\beta}(\{|u| > t\}) dt,$$

where $Ru = DI_1 u$ is the vector Riesz transform. When $\beta \in [1, d)$, we obtain that no such inequality is possible as a consequence of

Lemma 1.5. *Let $\beta \in [1, d)$. If Q denotes the cube $[0, 1]^d$, then*

$$\sup_{t>0} t \mathcal{H}_\infty^{d-\beta}(\{|DI_1 \chi_Q| > t\}) = +\infty.$$

Indeed, $\chi_Q \in L^1(\mathcal{H}_\infty^{d-\beta})$ for any $\beta \in (0, d)$, while the preceding shows that the weak-type quasi-norm of its Riesz transform is unbounded for $\beta \in [1, d)$ and so for such β one cannot have (1.15). This motivates

Open Problem 1.6. For $\beta \in (0, 1)$, can one find $u \in L^1(\mathcal{H}_\infty^{d-\beta})$ such that

$$\sup_{t>0} t \mathcal{H}_\infty^{d-\beta}(\{|DI_1 u| > t\}) = +\infty?$$

For $0 < s < d$, we use the normalization

$$\mathcal{H}_\infty^s(E) = \inf \left\{ \sum_i \omega_s r_i^s : E \subseteq \bigcup_i B(x_i, r_i) \right\}.$$

The harmless factor ω_s will always be absorbed into constants. The Choquet functional is

$$\|f\|_{L^1(\mathcal{H}_\infty^s)} = \int_0^\infty \mathcal{H}_\infty^s(\{|f| > t\}) dt,$$

and the space in the question is the completion of $C_c(\mathbb{R}^d)$ for this functional, as in [1].

Proof Intuition

A single cube has a logarithmic Riesz singularity near one hypersurface. If $s = d - \beta > d - 1$, a slab of thickness δ has s -content roughly $\delta^{1-\beta}$, so that one logarithmic singularity is too thin to force divergence.

Instead, place N very short intervals at spacing $1/N$. Immediately to the right of every interval the adjacent endpoint contributes $\asymp \log N$ to the Hilbert kernel, while all nonadjacent intervals contribute $o(1)$. The test gaps can be chosen so that their union carries a probability measure with uniform $(1 - \beta)$ -Frostman growth. Tensoring with a fixed $(d - 1)$ -dimensional cube lifts this to uniformly positive $(d - \beta)$ -content, and integration of the last Riesz kernel in the tangential variables produces the Hilbert kernel $1/h$ up to a bounded error.

Each finite comb has bounded Choquet cost but Riesz height $\asymp \log N$. Taking rapidly increasing N , summably decreasing amplitudes, and widely separated translates gives one function with finite Choquet norm and unbounded weak Riesz quasi-norm.

The Result

Theorem 1. *Let $d \geq 2$ and $0 < \beta < 1$. Put $s = d - \beta$. There exists a nonnegative function*

$$u \in L^1(\mathcal{H}_\infty^s) \cap L^1(\mathbb{R}^d)$$

such that

$$\sup_{t>0} t \mathcal{H}_\infty^s(\{x : |DI_1 u(x)| > t\}) = +\infty.$$

Moreover, u can be represented as a locally finite sum of nonnegative C_c^∞ functions.

Proof. Set

$$q = 1 - \beta \in (0, 1), \quad s = (d - 1) + q.$$

Choose real numbers

$$1 < r < p < \frac{1}{q}. \tag{1}$$

Step 1: a sparse one-dimensional comb. For a sufficiently large integer N , let

$$\ell_N = N^{-r}, \quad \delta_N = N^{-p},$$

and define

$$\begin{aligned} E_N &= \bigcup_{k=0}^{N-1} \left[\frac{k}{N}, \frac{k}{N} + \ell_N \right], \\ J_N &= \bigcup_{k=0}^{N-1} J_{k,N}, \\ J_{k,N} &= \left[\frac{k}{N} + \ell_N + \delta_N, \frac{k}{N} + \ell_N + 2\delta_N \right]. \end{aligned}$$

As $r, p > 1$, these intervals are disjoint and contained in $[0, 1]$ once N is large.

We claim that, uniformly for $z \in J_N$,

$$\left| \int_{E_N} \frac{dy}{z - y} \right| \geq c \log N. \quad (2)$$

Indeed, let $z \in J_{k,N}$ and write $z = k/N + \ell_N + \theta$ with $\delta_N \leq \theta \leq 2\delta_N$. The k th component contributes

$$\int_{k/N}^{k/N + \ell_N} \frac{dy}{z - y} = \log \frac{\ell_N + \theta}{\theta} \geq (p - r) \log N - \log 2.$$

For $j \neq k$, the distance from z to the j th component is at least $c|j - k|/N$. Therefore the absolute value of the sum of all nonadjacent contributions is at most

$$C \sum_{m=1}^N \frac{N\ell_N}{m} \leq CN^{1-r} \log N = o(1).$$

This proves (2).

Step 2: the target gaps have uniformly positive content. Let ν_N be normalized Lebesgue measure on J_N ; thus it has density $(N\delta_N)^{-1}$ on J_N and total mass one. We show that

$$\nu_N(I) \leq C|I|^q \quad (3)$$

for every interval $I \subset \mathbb{R}$, with C independent of N .

Write $\rho = |I|$. If $\rho \leq \delta_N$, then

$$\nu_N(I) \leq \frac{\rho}{N\delta_N} \leq \rho^q \frac{1}{N\delta_N^q} \leq \rho^q,$$

because $N\delta_N^q = N^{1-pq} \geq 1$. If $\delta_N \leq \rho \leq 1/N$, then I meets at most two components of J_N , and

$$\nu_N(I) \leq \frac{2}{N} \leq 2\delta_N^q \leq 2\rho^q.$$

If $1/N \leq \rho \leq 1$, then I meets at most $C(N\rho + 1)$ components, so $\nu_N(I) \leq C\rho \leq C\rho^q$. The case $\rho \geq 1$ is immediate. This proves (3).

Let

$$A = [-2, 2]^{d-1}, \quad A_0 = [-1, 1]^{d-1}, \quad B_N = A \times E_N, \quad F_N = A_0 \times J_N.$$

Take normalized $(d - 1)$ -dimensional Lebesgue measure λ on A_0 . For every ball $B(x, \rho) \subset \mathbb{R}^d$, the product probability measure $\mu_N = \lambda \otimes \nu_N$, supported on F_N , satisfies

$$\mu_N(B(x, \rho)) \leq C\rho^{d-1}\rho^q = C\rho^s \quad (0 < \rho \leq 1),$$

and the same estimate is trivial for $\rho \geq 1$. Consequently, for any ball cover $F_N \subseteq \bigcup_i B(x_i, r_i)$,

$$1 = \mu_N(F_N) \leq \sum_i \mu_N(B(x_i, r_i)) \leq C \sum_i r_i^s.$$

Taking the infimum over covers gives the uniform lower bound

$$\mathcal{H}_\infty^s(F_N) \geq c > 0. \quad (4)$$

Step 3: lift the Hilbert lower bound to a Riesz component. Up to a fixed nonzero dimensional constant, the last component of DI_1 is

$$R_d f(x) = c_d \text{ p. v. } \int_{\mathbb{R}^d} \frac{x_d - y_d}{|x - y|^{d+1}} f(y) dy.$$

For $x' \in A_0$ and $0 < |h| \leq 1$, set

$$K_{x'}(h) = c_d \int_A \frac{h}{(|x' - y'|^2 + h^2)^{(d+1)/2}} dy'.$$

Integration over all of $\mathbb{R}^{d-1}$ and the change of variables $w = |h|z$ give

$$c_d \int_{\mathbb{R}^{d-1}} \frac{h}{(|w|^2 + h^2)^{(d+1)/2}} dw = \frac{\kappa_d}{h}$$

for a nonzero constant κ_d . Since $x' \in A_0$, the translated set $x' - A$ contains the unit ball. Hence the omitted tail satisfies

$$\left| K_{x'}(h) - \frac{\kappa_d}{h} \right| \leq C|h| \int_{|w| \geq 1} |w|^{-d-1} dw \leq C|h|.$$

For $(x', z) \in F_N$, no principal value is needed and Fubini's theorem gives

$$\begin{aligned} R_d \mathbf{1}_{B_N}(x', z) &= \int_{E_N} K_{x'}(z - y_d) dy_d \\ &= \kappa_d \int_{E_N} \frac{dy_d}{z - y_d} + O(1), \end{aligned}$$

where the error is uniform because $z, y_d \in [0, 1]$ and $|E_N| = N\ell_N \leq 1$. By (2),

$$|R_d \mathbf{1}_{B_N}(x)| \geq c \log N - C \quad (x \in F_N). \quad (5)$$

We now smooth each block. Because $\text{dist}(F_N, B_N) = \delta_N$, choose a standard mollifier at a scale $\varepsilon_N < \delta_N/4$ so small that

$$v_N = \mathbf{1}_{B_N} * \rho_{\varepsilon_N} \quad \text{satisfies} \quad \sup_{x \in F_N} |R_d(v_N - \mathbf{1}_{B_N})(x)| \leq 1.$$

This is possible because $v_N \rightarrow \mathbf{1}_{B_N}$ in ordinary L^1 , while the Riesz kernel is uniformly bounded on pairs consisting of a point of F_N and a point in the ε_N -neighborhood of B_N . Thus $v_N \in C_c^\infty$, $0 \leq v_N \leq 1$, its support lies in a fixed bounded box independent of N , and

$$|R_d v_N(x)| \geq c \log N - C - 1 \quad (x \in F_N). \quad (6)$$

Step 4: glue the finite witnesses into one function. Let

$$a_n = 2^{-n}, \quad N_n = \lceil e^{4^n} \rceil, \quad R_n = L4^n,$$

where L will be chosen large. Increasing finitely many N_n if needed, assume all preceding estimates apply. Translate in the last coordinate:

$$\begin{aligned} \tilde{v}_n(x', x_d) &= v_{N_n}(x', x_d - R_n), \\ \tilde{F}_n &= F_{N_n} + (0, R_n). \end{aligned}$$

For large L , the supports of the $\tilde{v}_n$ are pairwise disjoint. Define

$$u = \sum_{n=1}^{\infty} a_n \tilde{v}_n.$$

First, u belongs to the precise Choquet space used in the question. The partial sums are in $C_c^\infty(\mathbb{R}^d)$. Since the supports are disjoint and every $\text{supp } \tilde{v}_n$ is contained in a translate of one fixed box, countable subadditivity of Hausdorff content yields, for $M' > M$,

$$\begin{aligned} & \int_0^\infty \mathcal{H}_\infty^s \left(\left\{ \left| \sum_{n=M+1}^{M'} a_n \tilde{v}_n \right| > t \right\} \right) dt \\ & \leq \sum_{n=M+1}^{M'} \int_0^\infty \mathcal{H}_\infty^s(\{a_n \tilde{v}_n > t\}) dt \leq C \sum_{n=M+1}^{M'} a_n \longrightarrow 0. \end{aligned}$$

Thus the partial sums are Cauchy in the defining functional, and their pointwise, locally finite sum u represents an element of $L^1(\mathcal{H}_\infty^s)$. Also $\|u\|_{L^1(\mathbb{R}^d)} \leq C \sum_n a_n < \infty$.

It remains to control interactions between different blocks. For $x \in \tilde{F}_n$ and $m \neq n$, the separation $R_j = L^j$, the kernel bound $|K(x - y)| \leq C|x - y|^{-d}$, and $\|\tilde{v}_m\|_1 \leq C$ give

$$\sum_{m \neq n} a_m |R_d \tilde{v}_m(x)| \leq CL^{-d}.$$

Indeed, all $m < n$ blocks are at distance comparable to R_n , while each $m > n$ block is at distance comparable to R_m ; summing the two geometric tails proves the displayed estimate. Choose L so large that this bound is at most one.

Using (6) and $\log N_n \geq 4^n$, we obtain, uniformly on $\tilde{F}_n$ and for all sufficiently large n ,

$$|R_d u(x)| \geq a_n(c \log N_n - C - 1) - 1 \geq c_1 2^n.$$

Since the norm of the vector Riesz transform dominates the absolute value of its last component, if $t_n = c_1 2^{n-1}$ then

$$\tilde{F}_n \subseteq \{|DI_1 u| > t_n\}.$$

Hausdorff content is translation invariant, so (4) implies

$$t_n \mathcal{H}_\infty^s(\{|DI_1 u| > t_n\}) \geq t_n \mathcal{H}_\infty^s(\tilde{F}_n) \geq ct_n \longrightarrow \infty.$$

This proves the theorem. □

Verification Notes

The proof is entirely quantitative. Its three threshold inequalities are $r > 1$, which makes the nonadjacent comb contribution vanish; $p > r$, which creates logarithmic height; and $pq < 1$, which ensures $N\delta_N^q \geq 1$ and hence positive target content. Such parameters exist exactly because $q = 1 - \beta < 1$.

The auxiliary script `code/check_comb_estimates.py` evaluates the exact finite sum

$$\sum_k \log \frac{|z - k/N|}{|z - k/N - \ell_N|}$$

on every target interval midpoint and samples the Frostman quotient $\nu_N(I)/|I|^q$ for representative values of q . This is a sanity check, not an input to the proof.

The constructed function is not compactly supported. This is permitted by the open problem, and its membership in the completion of C_c is verified directly by the Choquet-tail estimate above. The argument gives every $d \geq 2$ and every $0 < \beta < 1$, with no endpoint claim at $\beta = 0$ or $\beta = 1$.

Novelty Check

Before promotion, the run indexes and local source corpus were searched for the arXiv id, paper title, exact wording of Open Problem 1.6, and combinations of “Riesz transform”, “Hilbert transform”, “Hausdorff content”, and “Choquet”. Bounded web searches on 2026-08-09 used the same exact and close variants. They surfaced the original paper and related work on sparse operators in Choquet spaces [2], but no later answer to Open Problem 1.6 and no sparse-comb construction matching this one. This establishes only a bounded novelty check; expert bibliographic review remains necessary.

Human-Review Recommendation

Treat this as a likely-valid full solution requiring high-priority expert review. The most useful checks are the three-regime estimate (3), the tangential reduction to κ_d/h , and the passage from finite smooth blocks to the single Choquet-space function.

References

- [1] D. Spector, *A Noninequality for the Fractional Gradient*, Portugaliae Mathematica 76 (2019), no. 2, 153–168; arXiv:1906.05541.
- [2] N. Hatano, R. Kawasumi, H. Saito, and H. Tanaka, *Choquet integrals, Hausdorff content and sparse operators*, Archiv der Mathematik 124 (2025), 311–324; arXiv:2310.10135.